

COFFEYVILLE, KS, USA

WMO: 724519

Lat: 37.091N

Lon: 95.566W

Elev: 752

StdP: 14.30

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 93967

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	15.4	0.1	5.7	13.7	4.7	7.2	17.4	27.1	43.8	25.2	44.7	7.9	0	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	99.7	75.6	96.8	75.9	93.2	75.6	79.5	91.7	78.3	90.6	77.3	89.6	10.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.0	139.7	86.0	74.8	134.3	84.8	73.5	128.1	83.2	43.5	92.2	42.3	91.1	41.2	89.6	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.9	21.9	19.4	1.0	102.7	8.1	4.3	-4.8	105.8	-9.6	108.3	-14.1	110.7	-20.0	113.8		
			0.2	81.7	7.9	2.1	-5.5	83.1	-10.1	84.4	-14.5	85.5	-20.2	87.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.2	35.8	39.5	49.4	59.0	67.6	76.7	81.2	80.2	72.4	60.2	48.6	37.9	
	DBStd	18.23	10.76	11.31	10.74	8.73	7.51	5.56	5.36	5.86	7.18	9.05	9.65	10.34	
	HDD50	1491	453	327	140	22	1	0	0	0	0	18	138	392	
	HDD65	3947	906	715	491	216	62	2	0	1	17	203	495	839	
	CDD50	4834	13	32	123	294	547	800	967	937	671	335	97	18	
	CDD65	1812	0	1	9	37	142	352	502	472	238	55	4	0	
	CDH74	19577	1	9	76	342	1254	3643	5975	5394	2383	468	32	0	
	CDH80	9151	0	1	11	64	363	1564	3141	2852	1036	116	3	0	
Wind	WSAvg	8.8	9.3	9.6	10.7	10.9	8.9	8.5	7.3	6.9	7.0	8.2	9.2	9.0	
Precipitation	PrecAvg	41.9	1.5	1.6	3.0	4.4	6.4	5.8	3.7	3.2	4.1	3.3	2.6	2.2	
	PrecMax	57.9	4.3	4.4	7.4	10.8	11.3	16.3	8.1	8.3	9.2	8.5	7.4	5.9	
	PrecMin	23.6	0.1	0.0	0.3	0.7	2.3	1.6	0.8	0.0	0.9	0.2	0.0	0.2	
	PrecStd	7.6	1.0	1.3	2.0	2.2	2.7	3.3	1.9	1.8	2.0	2.2	2.0	1.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.8	74.4	81.0	86.1	90.4	97.4	104.0	104.7	99.7	88.0	78.4	69.9	
		MCWB	55.0	59.5	64.3	67.8	72.6	76.9	74.4	75.0	73.5	72.0	63.2	60.2	
	2%	DB	63.3	68.0	75.3	81.3	87.5	93.7	99.6	100.0	92.8	83.2	72.8	63.8	
		MCWB	52.5	55.9	61.1	65.4	72.3	76.1	76.4	75.4	73.9	68.1	60.1	55.4	
	5%	DB	58.6	63.3	71.5	78.1	84.1	90.9	96.7	97.1	90.0	79.3	69.2	58.7	
		MCWB	48.6	53.9	58.3	64.0	71.0	75.7	76.5	75.4	73.3	65.5	59.3	51.1	
	10%	DB	53.2	58.1	67.3	74.5	81.2	88.3	93.1	93.1	86.2	75.1	64.3	54.3	
		MCWB	45.1	49.3	56.9	62.6	70.1	74.7	76.5	74.9	71.6	64.2	55.6	47.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.7	62.9	66.7	70.8	76.4	80.3	81.6	80.7	78.1	74.0	66.4	62.9
			MCDB	63.9	70.3	77.3	81.8	85.6	92.3	92.1	92.3	89.8	84.0	73.9	68.0
2%		WB	54.6	59.2	63.9	68.5	74.6	78.5	79.7	78.8	76.4	71.6	63.1	57.8	
		MCDB	60.9	65.6	71.8	77.6	83.7	90.1	92.0	91.5	87.9	78.9	69.5	62.8	
5%		WB	49.5	55.0	61.3	66.6	72.9	77.3	78.4	77.6	75.0	69.2	60.4	52.9	
		MCDB	58.1	61.3	69.2	74.7	81.8	88.3	90.9	91.0	86.0	75.3	66.8	56.7	
10%		WB	45.4	49.6	58.3	64.4	71.2	75.9	77.6	76.4	73.5	66.4	57.2	47.8	
		MCDB	52.2	58.0	65.8	72.4	79.4	85.8	90.1	89.2	83.5	73.0	63.6	53.8	
Mean Daily Temperature Range	5% DB	MDBR	20.3	20.8	21.8	21.5	20.0	19.5	20.8	21.9	22.5	22.4	21.7	19.3	
		MCDBR	29.2	28.4	27.4	25.9	22.2	21.8	25.0	26.9	25.5	25.6	26.0	25.9	
		MCWBR	18.8	18.3	15.6	14.4	10.8	8.6	7.2	7.8	10.0	13.0	15.8	17.6	
	5% WB	MCDBR	25.2	25.2	21.4	21.1	19.2	19.5	20.5	22.0	21.1	19.7	22.1	21.0	
		MCWBR	18.1	18.1	14.0	13.7	10.8	9.1	8.0	8.4	9.2	12.0	15.7	17.2	
Clear-Sky Solar Irradiance	taub	0.300	0.311	0.353	0.384	0.422	0.427	0.442	0.446	0.390	0.349	0.312	0.293		
	taud	2.471	2.452	2.373	2.314	2.268	2.277	2.282	2.245	2.367	2.464	2.506	2.508		
	Ebn at Noon	284	294	289	284	274	272	266	263	273	276	277	279		
	Edh at Noon	26	30	36	40	43	43	42	43	36	30	25	23		
All-Sky Solar Radiation	RadAvg	777	982	1290	1627	1814	2082	2079	1868	1563	1153	834	661		
	RadStd	73	153	77	107	168	183	134	121	121	139	95	71		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page